STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-CHARACTERIZE-QUASI-COHERENT-BIS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1818-1837 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 1839-1860, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_algebraic-stacks-sheaves.json (pb=1148, stmt 1818-1837, proof 1839-1860).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma definition-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1751–1760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DU: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DT: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DX: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-restriction: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1185–1200. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DL: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Let $\mathcal{F}$ be a presheaf of modules on $\mathcal{X}$. The following are equivalent*

1. *$\mathcal{F}$ is an object of $Mod(\mathcal{X}_{Zar}, \mathcal{O}_\mathcal{X})$ and $\mathcal{F}$ is a quasi-coherent module on $(\mathcal{X}_{Zar}, \mathcal{O}_\mathcal{X})$ in the sense of Modules on Sites, Definition 03DL,*
2. *$\mathcal{F}$ is an object of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ and $\mathcal{F}$ is a quasi-coherent module on $(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ in the sense of Modules on Sites, Definition 03DL, and*
3. *$\mathcal{F}$ is a quasi-coherent module on $\mathcal{X}$ in the sense of Definition definition-quasi-coherent.*

Proof. Assume either (1), (2), or (3) holds. Let x be an object of $\mathcal{X}$ lying over the scheme U. Recall that $x^*\mathcal{F} = x^{-1}\mathcal{F}$ is just the restriction to $\mathcal{X}/x = (Sch/U)_\tau$ where $\tau = fppf$, $\tau = étale$, or $\tau = Zar$, see Section section-restriction. By the definition of quasi-coherent modules on a ringed site this restriction is quasi-coherent provided $\mathcal{F}$ is. By Descent, Proposition 03DX we see that $x^*\mathcal{F}$ is the sheaf associated to a quasi-coherent $\mathcal{O}_U$-module and is therefore a quasi-coherent module in the fppf, étale, and Zariski topology; here we also use Descent, Lemma 03DT and Definition 03DU. Since this holds for every object x of $\mathcal{X}$, we see that $\mathcal{F}$ is a sheaf in any of the three topologies. Moreover, we find that $\mathcal{F}$ is quasi-coherent in any of the three topologies directly from the definition of being quasi-coherent and the fact that x is an arbitrary object of $\mathcal{X}$. □